

Argus-Unified: Towards A Compact and Economical Unified Model for Image Understanding and Generation

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Abstract

Unifying visual understanding and generation in one model holds immense promise, but remains challenging and expensive due to heavy compute and data demands and conflicts between the visual features needed for these two capabilities. To address these challenges, we present Argus-Unified, a compact, effective and unified multimodal model built with low demand on computation and data. Instead of aligning modalities from scratch, Argus-Unified effectively leverages pretrained vision-language models (VLMs) that provide strong multimodal priors. Specifically, we introduce hybrid visual tokens that preserve continuous tokens for understanding while learning discrete tokens for generation from a frozen unified vision encoder. Our training pipeline includes two stages: the first stage learns a quantizer and image decoder on top of the frozen vision encoder, the second stage trains the LLM initialized from a pretrained VLM for the unified multimodal modeling. Using by far the least amount of data (15.6M) and the lowest cost ($\sim \$2,000$), we demonstrate that unified multimodal models can be trained economically while achieving strong performance in both understanding and generation. Notably, our model attains state-of-the-art multimodal understanding on GQA, POPE, and VQAv2, and competitive generation quality compared to models with dedicated vision encoders (e.g., Janus, Janus-Pro), all at $\sim 10\times$ lower cost and with $\sim 5\times$ less data. We envision Argus-Unified as a useful baseline that lowers the development barrier for unified models.

1 Introduction

With the success of large language models (LLMs) that unify natural language tasks (Touvron et al., 2023; Achiam et al., 2023; Yang et al., 2025a), the pursuit at the forefront of unified multimodal models (UMM) that further integrates visual un-

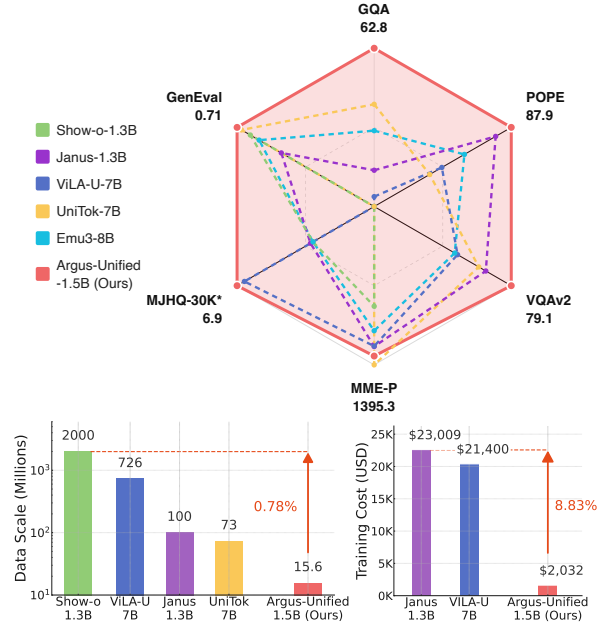


Figure 1: Argus-Unified achieves strong performance on both image understanding and generation across six benchmarks with only $\sim \$2,000$ economical cost and 15.6M data, substantially lower than existing unified multimodal models (UMMs). * denotes values inverted for illustration, where the original metric is lower-better.

derstanding and generation have become increasingly promising (Team, 2024; Wang et al., 2024; Wu et al., 2024c,a; Ma et al., 2025a). Companies and institutions have built vision-language models (VLMs) for visual understanding based on pretrained LLMs (Chen et al., 2024d; Lu et al., 2024a; Wu et al., 2024d; Marafioti et al., 2025; Bai et al., 2025; Zhu et al., 2025). Meanwhile, pioneering researches on visual generation has adopted autoregressive (AR) paradigm that predicts next visual tokens (Li et al., 2024c; Sun et al., 2024; Tian et al., 2024). These advances enable the unification of visual understanding and generation based on the AR paradigm. However, existing UMMs primarily align modalities from scratch—relying on independently pretrained LLM and vision en-

coder—without harnessing the multimodal priors in VLMs (Wu et al., 2024c,a; Ma et al., 2025a).

Leveraging VLMs for unified multimodal models is not straightforward due to the tokenization difference between understanding and generation. VLMs use *continuous visual tokens* (i.e., visual features) from the vision encoder which contain rich semantic information (Fig. 2(a)) (Liu et al., 2023, 2024b; Zhu et al., 2025; Bai et al., 2025). In contrast, autoregressive generation models commonly use *discrete visual tokens* based on VQVAE (Van Den Oord et al., 2017) or VQGAN (Esser et al., 2021a). To address this mismatch, prior autoregressive UMMs mainly follow two approaches. Some works (Wang et al., 2024; Wu et al., 2024a,c; Ma et al., 2025a) adopt discrete tokens for both understanding and generation (Fig. 2(b)), at the cost of losing fine-grained semantic information critical for the understanding ability (Wu et al., 2024c; Ma et al., 2025a). The others (Wu et al., 2025a; Chen et al., 2025b; Jiao et al., 2025) use two vision encoders, one for understanding and one for generation (Fig. 2(c)). They use continuous tokens for the understanding and discrete tokens from a separate vision tokenizer to fit the generation task. However, this design increases model parameters with an additional encoder.

In light of this, we propose Argus-Unified, a compact unified multimodal model that effectively leverages the pretrained VLMs to provide strong vision-language priors. Argus-Unified employs task-specific visual tokens from a single vision encoder: continuous tokens for understanding and discrete tokens for generation, as shown in Fig. 2(d). This *hybrid token* design enables easy transformation of a pretrained VLM into a UMM with a simple two-stage training:

(i) *Unified Vision Tokenizer Training*: Argus-Unified trains a quantizer to discretize visual tokens and an image decoder to generate images. The vision encoder is based on the *frozen* pretrained VLM. The continuous tokens from the VLM are for understanding and the discrete tokens from the quantizer are for generation. With the encoder frozen, the vision tokenizer can be trained solely with image reconstruction loss, without requiring vision-language alignment (Ma et al., 2025a).

(ii) *Unified Multimodal Training*: Argus-Unified optimizes the LLM for both image understanding and generation. Instead of training from pretrained LLMs like prior UMMs (Wu et al., 2025a; Chen et al., 2025b; Wu et al., 2024c), we initialize the

LLM from a pretrained VLM, enabling the model to start from the learned vision-language alignments rather than learning them from scratch.

We conduct a comprehensive evaluation on both multimodal understanding and image generation benchmarks, and compare Argus-Unified with the latest UMMs, including Emu3 (Wang et al., 2024), Janus series (Wu et al., 2025a; Chen et al., 2025b), Show-o series (Xie et al., 2024, 2025), VILA-U (Wu et al., 2024c), and more. Argus-Unified achieves the best performance on multimodal understanding benchmarks such as GQA (Hudson and Manning, 2019), POPE (Li et al., 2023), and VQAv2 (Antol et al., 2015; Goyal et al., 2017) and competitive performance on generation benchmarks, with a total cost of about \$2,000.¹ Our Argus-Unified-0.5B, one of the smallest UMMs using only 0.5B LLM, surpasses methods like Emu3 (Wang et al., 2024) and Chameleon (Team, 2024) with over 7B LLMs. Notably, we use only 15.6M training data, significantly less than the existing works that use tens of millions to hundreds of millions, even billions of data (as shown in Fig. 1).

2 Related Work

2.1 Vision-language Models

Vision-language models (VLMs) are multimodal models that process both visual and textual inputs (Radford et al., 2021; Liu et al., 2023; Huang et al., 2026). Pioneering works such as CLIP (Radford et al., 2021) employ contrastive learning which takes natural language as supervision for visual representation learning (Zhuang et al., 2025). Recent VLMs, including LLaVA series (Liu et al., 2023, 2024b; Zhu et al., 2024), InternVL series (Chen et al., 2024d; Zhu et al., 2025), Qwen-VL series (Bai et al., 2023, 2025), commonly integrate a vision encoder with LLMs via a connector, e.g. a multi-layer perceptron (MLP). Fig. 2(a) illustrates the high-level architecture of these VLMs. The vision encoder—usually a Vision Transformer (ViT) (Dosovitskiy, 2020) such as CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or InternViT (Chen et al., 2024e)—produces continuous visual tokens that are then aligned with textual embeddings. Despite the advances of VLMs, they remain limited to text-based visual understanding and lack the ability of visual generation.

¹We estimate the cost of an 8×H100 GPU node at \$20.3/hour and an 8×A100 GPU node at \$8.56/hour according to <https://cloud-gpus.com/>.

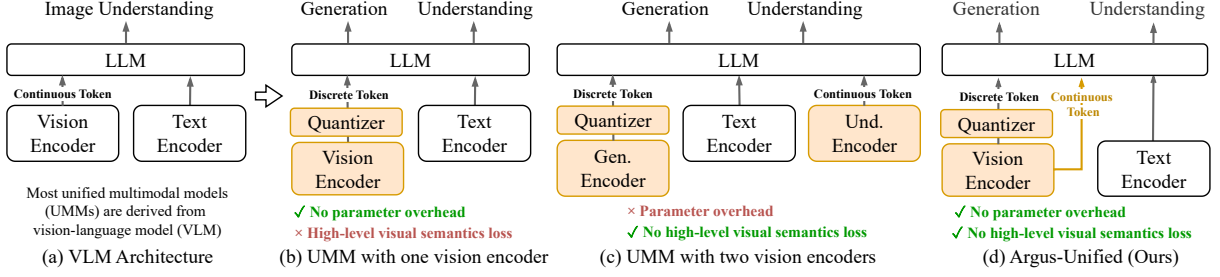


Figure 2: Illustrations of model architectures of autoregressive vision-language model (VLM) and unified multimodal models (UMMs). (a) Widely adopted VLM architecture that uses continuous tokens (i.e., features) for image understanding. (b) Commonly adopted UMM architecture that uses a quantizer to produce discrete tokens for both understanding and generation based on a vision encoder (Wu et al., 2024c; Ma et al., 2025a; Team, 2024; Wang et al., 2024; Wu et al., 2024a). (c) Another popular UMM architecture that uses a generation (gen.) vision encoder with quantizer to produce discrete tokens and an understanding (und.) vision encoder for continuous tokens (Wu et al., 2025a; Chen et al., 2025b; Qu et al., 2025; Jiao et al., 2025). (d) We propose hybrid tokens, a new design for UMMs that leverages the best token types for respective tasks — continuous tokens for understanding and discrete tokens for generation, while keeping the architecture lightweight.

2.2 Unified Multimodal Models

Unified multimodal models (UMMs) aim to integrate both visual understanding and generation in a single model (Lu et al., 2024b; Wu et al., 2024b; Wang et al., 2024; Team, 2024). Existing UMMs can be broadly categorized into three designs:

Autoregressive-based UMMs are generally built upon autoregressive (AR) LLMs and jointly model image and text generation. Some works replace the vision encoder with that of a VQVAE (Van Den Oord et al., 2017) or VQGAN (Esser et al., 2021b) like tokenizer (Fig. 2(b)) (Wang et al., 2024; Team, 2024; Wu et al., 2024c; Ma et al., 2025a; Han et al., 2025; Wu et al., 2024a), while others incorporate an additional vision tokenizer dedicated for image generation (Fig. 2(c)) (Wu et al., 2025a; Chen et al., 2025b; Jiao et al., 2025; Qu et al., 2025).

Diffusion-based UMMs employ diffusion models (Rombach et al., 2022; Podell et al., 2023; Sehwag et al., 2025) instead of LLMs to generate images and texts (Yang et al., 2025b; Li et al., 2025b; Shi et al., 2025). However, they remain in early stages with suboptimal performance, partly due to the difficulty of mapping discrete text into a continuous space and capturing sequential dependencies.

Hybrid AR+Diffusion UMMs combine AR for text and diffusion for image generation. Some works assemble a separate diffusion model on top of a pretrained VLM (Tong et al., 2024; Pan et al., 2025), forming loosely coupled architecture and separate token spaces rather than a fully unified design. Others integrate both paradigms within a transformer (Ma et al., 2025b; Xie et al., 2024,

2025; Wu et al., 2025b; Fan et al., 2025). While diffusion offers high image fidelity, its independent training paradigm limits reuse of visual priors from pretrained AR-based VLMs.

Our Argus-Unified is *AR-based UMM* that effectively leverages pretrained VLMs and introduces a hybrid token design (Fig. 2(d)), achieving the state-of-the-art performance in this paradigm. Compared with other paradigms, it attains superior multimodal understanding performance and competitive generation quality with substantially lower training data and cost.

3 Methodology

In this section, we present the design and training strategy of Argus-Unified, a compact unified multimodal model (UMM) that performs both image understanding and image generation by effectively leveraging pretrained vision-language models (VLMs). Argus-Unified introduces a hybrid token design from a single vision encoder, which outputs both continuous tokens rich in semantics for understanding and quantized discrete tokens suitable for generation. This design leverages the benefits of both token types within one encoder while avoiding parameter overhead.

Fig. 3 provides an overview of the Argus-Unified. It consists of two major stages: (i) Unified Vision Tokenizer Training and (ii) Unified Multimodal Training. In Stage 1 (Fig. 3(a)), Argus-Unified builds a unified vision encoder by training a quantizer and decoder on top of the frozen vision encoder from a pretrained VLM. It enables the vision encoder to produce both continuous tokens

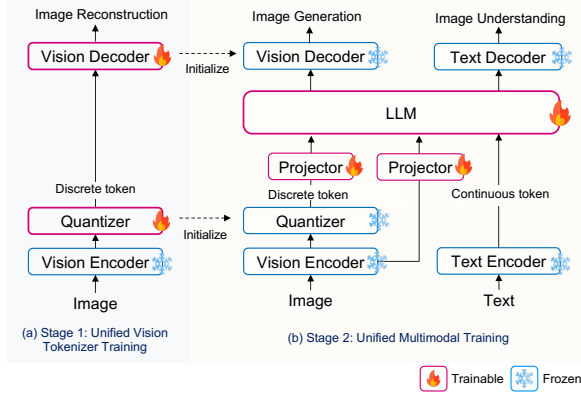


Figure 3: Overview of Argus-Unified. Argus-Unified transforms pretrained VLMs into UMMs via a novel hybrid token design and a two-stage training framework. (a) In Stage 1, a quantizer and vision decoder are trained on top of the frozen vision encoder to enable it to produce both continuous tokens for understanding and discrete tokens for generation. (b) In Stage 2, we initialize the model with pretrained VLM components and quantizer from Stage 1, then perform unified multimodal training to jointly optimize the LLM and projectors for understanding and generation.

and discrete tokens for the next stage. In Stage 2 (Fig. 3(b)), we initialize the model with pretrained VLM components and quantizer from Stage 1, then perform unified multimodal training to jointly optimize the LLM for both image understanding and generation. Now, we describe these two training stages in detail.

3.1 Unified Vision Tokenizer (Stage 1)

We employ a shared vision encoder for image understanding and generation for architecture simplicity and model compactness. Unlike prior UMMs that fine-tune the vision encoder (Wu et al., 2024c; Ma et al., 2025a; Wu et al., 2025a; Chen et al., 2025b), we *freeze* a pretrained vision encoder from a strong VLM, preserving its vision-language alignments for understanding while efficiently extending it to image generation. In comparison, prior UMMs with a unified vision encoder do not freeze their vision encoders (Wang et al., 2024; Wu et al., 2024c; Ma et al., 2025a), as they need to simultaneously optimize discrete tokens for generation and align visual representations with texts.

The key to enabling autoregressive image generation lies in producing discrete visual tokens from the vision encoder. We adopt the widely used VQVAE (Van Den Oord et al., 2017) and VQGAN (Esser et al., 2021b) architecture and trains a quantizer and an image decoder on top of the

frozen pretrained vision encoder, as depicted in Fig. 3(a). The quantizer discretizes the continuous embeddings from the vision encoder by mapping them to the nearest entries in a learned codebook of latent vectors, thus producing discrete codes that can be used as the prediction labels in autoregressive LLMs. The decoder is trained to reconstruct the image from these discrete codes.

We train the quantizer and decoder with the following loss:

$$\mathcal{L} = \mathcal{L}_R + \lambda_P \mathcal{L}_P + \lambda_G \mathcal{L}_G + \lambda_{VQ} \mathcal{L}_{VQ}, \quad (1)$$

where $\mathcal{L}_R$ is the pixel-wise reconstruction loss, $\mathcal{L}_P$ is the perceptual loss using LPIPS (Zhang et al., 2018), $\mathcal{L}_G$ is the adversarial loss for enhancing reconstruction fidelity (Karras et al., 2019), $\mathcal{L}_{VQ}$ is the vector quantization loss that minimizes distance between the vision encoder output and its nearest codebook entry (Van Den Oord et al., 2017), and λ s are the weight coefficients for corresponding loss terms. With the vision encoder frozen, our tokenizer is trained without the image-text loss (Radford et al., 2021) used in prior works (Wu et al., 2024c; Ma et al., 2025a). This not only simplifies training—requiring only unlabeled image data—but also reduces computational costs.

3.2 Unified Multimodal Training (Stage 2)

Fig. 3(b) illustrates the overall architecture of our Argus-Unified. It consists of three main components: (1) the vision encoder, LLM, text encoder, and text decoder initialized from the pretrained VLM; (2) the quantizer and image decoder initialized from Stage 1; and (3) two randomly initialized task-specific projectors for understanding and generation, respectively. In this stage, the LLM and the two projectors are trainable.

Hybrid Token Design. Argus-Unified introduces hybrid tokens based on a unified vision encoder: continuous tokens for understanding and discrete visual tokens for generation. Starting from a pretrained VLM vision encoder, we use its continuous visual representations for understanding, followed by a quantizer to produce discrete visual tokens for generation. Both token types are projected into a shared LLM embedding space through separate projectors, enabling the model to handle modalities coherently. Moreover, this design is cohesive with *freezing* vision encoder in Stage 1, allowing Argus-Unified to fully leverage pretrained VLMs while extending them for image generation.

Prior UMMs mostly adopt discrete visual codes from VQGAN-like vision tokenizers (Wu et al., 2024c; Ma et al., 2025a) for both tasks, which limits the semantic richness for understanding.

Training Objective. We train the LLM using the standard next-token prediction objective (Radford et al., 2019; Brown et al., 2020). Each sample is represented as a multimodal token sequence formed by concatenating visual tokens from the vision encoder and text tokens from the text tokenizer. Specifically, the vision encoder converts an image into a 1D token sequence wrapped with special tokens `<image_start>` and `<image_end>`, inserted among text tokens following the format of chat template. These multimodal tokens are fed into the LLM, producing text tokens for understanding and visual tokens for generation. For image understanding, the model minimizes the negative log-likelihood (NLL) over text tokens. For image generation, the output visual tokens are processed by a vision head and subsequently used to compute the NLL loss, similar to VILA-U (Wu et al., 2024c). During inference, text tokens are decoded by the text decoder to generate text, while visual tokens are passed to the image decoder to generate image.

Training Recipes. Our unified multimodal training consists of pretraining and supervised fine-tuning (SFT), designed to effectively adapt a pre-trained VLM into a UMM. Prior UMMs typically require large-scale data for multimodal pretraining to teach the LLM visual token modeling, *i.e.*, vision-language alignment (Wu et al., 2024c,a). Conversely, our LLM is initialized from a pre-trained VLM and already exhibits strong image understanding and text generation abilities. Therefore, we can warm up the LLM for visual token generation with substantially less data during pretraining. Specifically, we use generation data JourneyDB (4.1M) (Sun et al., 2023) and a smaller scale understanding data ShareGPT4V (1.25M) (Chen et al., 2024c) to maintain modality alignment. The total data amount in pretraining is significantly lower than prior works (Wu et al., 2024a; Ma et al., 2025a; Wu et al., 2025b), which use over 60M data.

In SFT, we further fine-tune the LLM on roughly balanced data of understanding and generation. For image understanding, we use WiT (Srinivasan et al., 2021), ShareGPT4V (Chen et al., 2024c),

VFLAN (Chen et al., 2024a), ScienceQA (Lu et al., 2022), and MGM-Instruct (Li et al., 2024d), totaling 5.47M samples. For text-to-image generation, we use JourneyDB (Sun et al., 2023), BLIP3-o (Chen et al., 2025a), and Echo-4o Instruct (Ye et al., 2025), totaling 4.23M image-text pairs.

Altogether, we curate 9.7M image-text pairs for unified multimodal training. We find that in our setting of leveraging pretrained VLMs, expanding pretraining data beyond a moderate size offers diminishing or even negative returns, while enlarging SFT data can improve performance. More details are discussed in Sec. 4.3.

4 Experiments

In this section, we detail the experimental setup, compare Argus-Unified with existing UMMs, VLMs, and image generation models, and present comprehensive ablation studies.

4.1 Experiment Setup

Datasets. We curated a total number of 15.6 million public data, consisting of 9.7M image-text pairs for Stage 2 (as discussed in Sec. 3.2) and additional 5.9M images from CC3M (Sharma et al., 2018), DALL-E 3 (Egan et al., 2024), and DiffusionDB (Wang et al., 2022). Stage 1 training uses these images and the images from the image-text dataset, resulting in 14M images for vision tokenizer training. Note that we only utilize publicly available datasets without using any internal or proprietary data like prior works (Wu et al., 2024c; Ma et al., 2025a) to facilitate reproducibility. As shown in Fig. 1 and Tab. 1, our total data volume is an order of magnitude smaller than that of existing UMMs, some of which require hundreds of millions to billions of samples.

Evaluation Metrics. We evaluate the multimodal understanding capability of Argus-Unified on standard benchmarks, including POPE (Li et al., 2023), VQAv2 (Antol et al., 2015; Goyal et al., 2017), GQA (Hudson and Manning, 2019), and MME (Zhang et al., 2021). We also assess the image generation capability on two complementary benchmarks: MJHQ-30K (Li et al., 2024a) and GenEval (Ghosh et al., 2023). MJHQ-30K evaluates visual fidelity using the Fréchet Inception Distance (FID) computed over 30K images. GenEval focuses on compositional reasoning and object consistency in generated images.

Implementation Details. We focus on com-

¹For the works that do not report data scale, we tried our best to estimate it from the description of datasets used.

Models	Type	LLM Scale	Data Scale ²	Image Understanding				Image Generation	
				GQA $\uparrow$	MME-P $\uparrow$	POPE $\uparrow$	VQAv2 $\uparrow$	MJHQ-30K $\downarrow$	GenEval $\uparrow$
MetaQuery (Pan et al., 2025)	AR+Diff.	0.5B	27.4M	-	1238.0	-	-	6.3	0.74
UniFork (Li et al., 2025a)	AR	0.76B	82.3M	55.1	1208.0	85.8	70.0	10.6	0.46
Harmon (Wu et al., 2025b)	AR+MAR	0.5B	109M	56.3	1148.0	86.5	-	6.1	0.71
Argus-Unified (Ours)	AR	0.5B	15.6M	61.4	1312.6	87.5	77.3	8.0	0.66
MMaDA (Yang et al., 2025b)	Diffusion	8B	2B+	61.3	1410.7	86.1	76.7	-	0.63
Seed-X (Ge et al., 2024)	AR+Diff.	13B	158M+	49.1	1457.0	84.1	71.2	-	0.51
LaViT (Jin et al., 2023)	AR+Diff.	7B	293M	46.8	-	-	66.0	-	-
ILLUME+	AR+Diff.	3B	73M	-	1414.0	87.6	-	6.0	0.53
Chameleon (Team, 2024)	AR	34B	1.4B	-	604.5	-	69.6	-	-
Chameleon (Team, 2024)	AR	7B	1.4B	-	202.7	-	-	-	0.39
TokenFlow-L (Qu et al., 2025)	AR	13B	760M+	62.6	1365.4	85.0	73.9	-	-
Emu3 (Wang et al., 2024)	AR	8B	-	60.3	1243.8	85.2	75.1	-	0.66
TokLIP-L (Lin et al., 2025)	AR	7B	125M	59.5	1488.4	84.1	-	-	-
LWM (Liu et al., 2024a)	AR	7B	1B	44.8	-	75.2	55.8	17.8	0.47
SemHiTok-256 (Chen et al., 2025c)	AR	7B	70M	60.3	1449.0	83.4	-	5.4	-
UniTok (Ma et al., 2025a)	AR	7B	1B+	61.1	1448.0	83.2	76.8	7.5	-
Liquid (Wu et al., 2024a)	AR	7B	60M	58.4	1119.3	81.1	71.3	5.5	-
ViLA-U-256 (Wu et al., 2024c)	AR	7B	726M	58.3	1336.2	83.9	75.3	12.8	0.41
D-Dit-512 (Li et al., 2025b)	Diffusion	2B	40M	59.2	1124.7	84.0	60.1	-	0.50
Show-o2 (Xie et al., 2025)	AR+Diff.	1.5B	66M	60.0	1450.9	-	-	-	0.73
Show-o-512 (Xie et al., 2024)	AR+Diff.	1.3B	2B+	58.0	1097.2	80.0	69.4	-	0.68
Harmon (Wu et al., 2025b)	AR+MAR	1.5B	109M	58.9	1155.0	87.6	-	5.2	0.76
Janus-Pro (Chen et al., 2025b)	AR	1.5B	234M	59.3	1444.0	86.2	-	-	0.73
Janus (Wu et al., 2025a)	AR	1.3B	100M+	59.1	1338.0	87.0	77.3	10.1	0.61
Argus-Unified (Ours)	AR	1.5B	15.6M	62.8	1395.3	87.9	79.1	6.9	0.71

Table 1: Performance comparison to the leading unified multimodal models (UMMs) on multimodal understanding and image generation benchmarks. Our model achieves the state-of-the-art performance among autoregressive (AR) models, excelling on both understanding and generation. Compared with diffusion-based and hybrid (AR+MAR or AR+Diffusion) UMMs, Argus-Unified achieves the best multimodal understanding results on GQA, POPE, and VQAv2, while maintaining competitive generation performance with significantly less data.

pact unified multimodal models and develop two variants: Argus-Unified-0.5B and Argus-Unified-1.5B. Argus-Unified-0.5B is built on InternVL3-1B (Zhu et al., 2025), consisting of Qwen2.5-0.5B (Qwen et al., 2024) as the LLM and an InternViT-300M (Chen et al., 2024e) as the vision encoder. Argus-Unified-1.5B is built on InternVL3-2B (Zhu et al., 2025), consisting of Qwen2.5-1.5B (Qwen et al., 2024) as the LLM and InternViT-300M (Chen et al., 2024e) as the vision encoder. For generation, we resize images to 448×448 to use the same vision encoder and adopt multi-codebook design for quantization inspired by UniTok (Ma et al., 2025a). The image decoder is initialized from ViTamin-L/16 (Chen et al., 2024b). Both understanding and generation projectors are single-layer MLPs.

4.2 Performance Comparison

We compare Argus-Unified with over 20 latest unified multimodal models (UMMs) with LLM scales from 0.5B to 34B in Tab. 1. Existing UMMs generally follow one of three design paradigms: (1) autoregressive (AR) for both understanding and generation, e.g., Emu3 (Wang et al., 2024), Janus (Wu

Models	LLM Scale	Vision Enc. Scale	Data Scale	GPU Hours
Janus-Pro (Chen et al., 2025b)	1.5B	345M	234M	$27648 \times \text{A100}$
Janus (Wu et al., 2025a)	1.3B	345M	100M+	$21504 \times \text{A100}$
ViLA-U (Wu et al., 2024c)	7B	316M	726M	$20000 \times \text{A100}$
Harmon (Wu et al., 2025b)	1.5B	943M	109M	$6144 \times \text{A100}$
Argus-Unified	1.5B	304M	15.6M	$800 \times \text{H100}$

Table 2: Compute comparison with unified multimodal models. Argus-Unified is trained with $8 \times \text{NVIDIA H100}$ for 4.17 days (Stage 1 + Stage 2), requiring substantially less compute.

et al., 2025a), and ViLA-U (Wu et al., 2024c); (2) diffusion-based for both understanding and generation (Diffusion), such as MMaDA (Yang et al., 2025b) and D-Dit (Li et al., 2025c); or (3) hybrid of AR and Diffusion or MAR (Li et al., 2024b) (AR+Diff. or AR+MAR), which uses AR for understanding and diffusion or MAR modeling for generation—either by assembling an external diffusion model (e.g., MetaQuery (Pan et al., 2025)) or fusing diffusion or MAR modeling into AR (e.g., Show-o (Xie et al., 2024, 2025)).

Comparison with Unified Multimodal Models. Argus-Unified demonstrates strong performance across both understanding and generation benchmarks compared with existing UMMs. Among AR

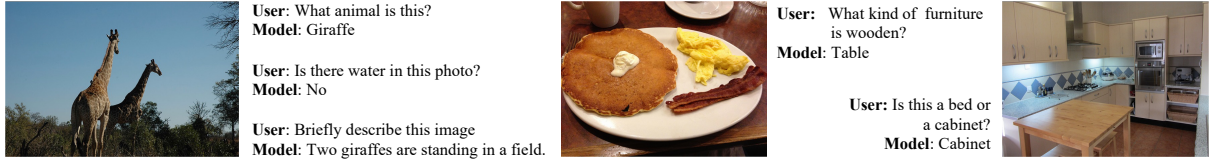


Figure 4: Image understanding examples produced by Argus-Unified-1.5B.

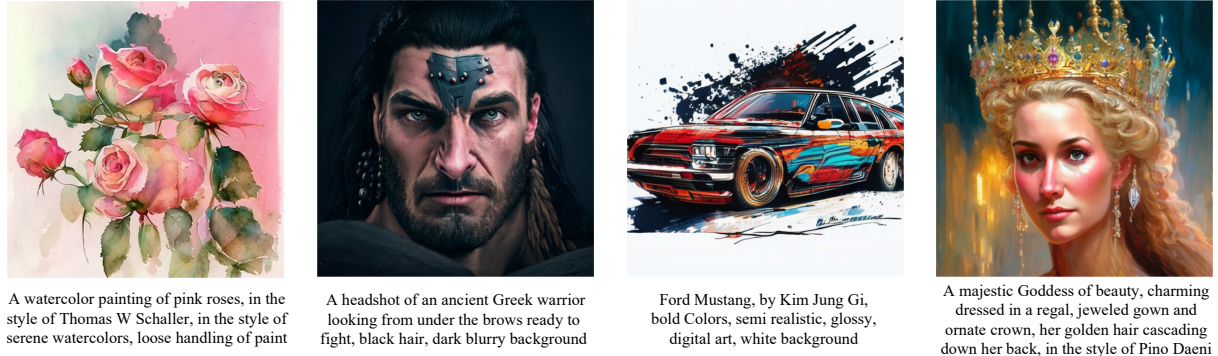


Figure 5: Images generated by Argus-Unified-1.5B. More examples are provided in the Fig. 8.

UMMs, Argus-Unified achieves the state-of-the-art performance, surpassing representative works like Janus (Wu et al., 2025a) (which has a specialized encoder for generation) and VILA-U (Wu et al., 2024c) across all benchmarks, and outperforming the other AR UMMs on majority of the benchmarks. Notably, Argus-Unified-1.5B even outperforms models with substantially larger LLMs (7B–13B) using far less data (only 15.6M).

Compared with diffusion-based and hybrid (AR+MAR or AR+Diffusion) UMMs, Argus-Unified-0.5B achieves the best results on all image understanding benchmarks among models with LLMs under 1B parameters and Argus-Unified-1.5B attains the best performance on GQA, POPE, and VQAv2, while remaining competitive on MME-P. For image generation, Argus-Unified-1.5B outperforms models with diffusion modeling such as Seed-X (Ge et al., 2024) and Show-o (Xie et al., 2024), and achieves comparable results with Harmon (Wu et al., 2025b) and MetaQuery (Pan et al., 2025). Note that MetaQuery uses the same 0.5B-scale LLM but assembles an external diffusion model with approximately 1.6B parameters. Although Harmon-1.5B uses the same LLM scale as ours, its vision encoder with 943M parameters is about $3\times$ larger than ours and it uses about $7\times$ more data and more computation, as shown in Sec. 4.1. We also provide qualitative examples of understanding and generation in Fig. 4 and Fig. 5. More comparisons with models that only support image understanding or generation are provided in

w/ CFG	Image Understanding				Image Generation	
	GQA↑	MME-P↑	POPE↑	VQAv2↑	MJHQ↓	GenEval↑
×	63.0	1405.1	87.7	79.1	8.2	0.71
✓	62.8	1395.3	87.9	79.1	6.9	0.71

Table 3: Performance of Argus-Unified training with (w/) and without classifier-free guidance (CFG) (Ho and Salimans, 2022). CFG improves MJHQ-30K (Li et al., 2024a) FID while preserving performance on the others.

Tabs. 8 and 9 in the Supplementary.

4.3 Recipes and Ablation Studies

Impact of Classifier-Free Guidance (CFG). CFG is widely used to improve generation quality and control how closely output follows a text prompt (Ho and Salimans, 2022). We empirically study the impact of CFG in unified multimodal training and present the results in Tab. 3. Using CFG in training effectively improves the FID score on MJHQ-30K (Li et al., 2024a) while preserving the performance on GenEval (Ghosh et al., 2023) and understanding benchmarks. These results demonstrate the effectiveness of CFG in the unified multimodal training. For the rest of the ablation studies, we do not apply CFG to disentangle the impact of different components.

Vision Tokenizer. We evaluate our vision tokenizer on ImageNet (Deng et al., 2009) reconstruction FID (rFID) and compare it with recent tokenizers in Tab. 4. Despite the vision encoder is frozen and using only 14M images, ours achieves comparable ImageNet rFID to tokenizers designed for

Type	Gen. Only			Unified Models				
Models	VQGAN	RQ-VAE	VAR	TokenFlow	VILA-U	Argus-Unified	Argus-Unified (Codebook×2)	Argus-Unified (SigLIP2)
Data Scale	-	-	-	-	700M	14M	14M	14M
ImageNet rFID ↓	4.98	1.30	0.90	1.37	1.80	1.60	1.04	0.63

Table 4: Comparison of vision tokenizers on ImageNet (Deng et al., 2009) reconstruction FID (rFID). Argus-Unified achieves competitive performance with a frozen vision encoder and only 14M data. A larger codebook size (×2) and alternative vision encoders (e.g., SigLIP2-Large (Tschannen et al., 2025)) can further improve results.

Pretrained Weights	Image Understanding				Image Generation	
	GQA ↑	MME-P ↑	POPE ↑	VQAv2 ↑	MJHQ-30K ↓	GenEval ↑
Pretrained ViT & LLM	56.54	1361.51	85.03	75.26	8.77	0.61
Pretrained VLM	62.95	1405.06	87.70	79.10	8.22	0.71

Table 5: Impact of initialization strategy on unified multimodal training. Using pretrained VLM (InternVL3-2B) consistently outperforms initializing from separate pretrained ViT (InternViT) and LLM (Qwen2.5-1.5B).

Token Type	Image Understanding				Image Generation	
	GQA ↑	MME-P ↑	POPE ↑	VQAv2 ↑	MJHQ-30K ↓	GenEval ↑
Discrete Token	60.10	1257.43	85.73	73.23	8.34	0.72
Hybrid Token	62.95	1405.06	87.70	79.10	8.22	0.71

Table 6: Comparison of discrete vs. hybrid token designs. Using continuous tokens for understanding and discrete tokens for generation (hybrid token) significantly enhances image understanding while preserving generation quality.

generation-only or unified models. We further visualize the reconstructed images in Fig. 7. Notably, Argus-Unified with a frozen encoder significantly outperforms VILA-U (Wu et al., 2024c), which does not freeze the encoder.

In addition, we conduct ablations by doubling the codebook size and replacing the encoder with a pretrained SigLIP2-Large (Tschannen et al., 2025). Both modifications lead to improved rFID and reconstruction quality. These results indicate that neither the tokenizer nor freezing the encoder freezing constitutes a bottleneck for unified models. Instead, our findings highlight the importance of effective training strategies in unified modeling, including classifier-free guidance (CFG), initialization from pretrained vision-language models (VLMs), and careful dataset design. We discuss these factors in greater detail in the subsequent ablation studies.

Pretrained VLM vs. Pretrained ViT & LLM. Tab. 5 compares initializing the model from a pretrained VLM (InternVL3-2B) versus separately pretrained vision encoder (InternViT) and language model (Qwen2.5-1.5B). Note that InternVL3-2B is built upon InternViT (Chen et al., 2024e) and Qwen2.5-1.5B (Qwen et al., 2024), i.e., the architectures are identical across settings. We train both using the exact same pipeline and data. The results show that initializing from a pretrained VLM yields

substantially stronger performance, demonstrating the benefit of leveraging pretrained VLMs.

Token Type. Most existing UMMs use discrete tokens for both understanding and generation. In contrast, we introduce a hybrid token design that uses continuous tokens for understanding and discrete tokens for generation. As shown in Tab. 6, this hybrid design substantially enhances image understanding performance while preserving competitive results on image generation benchmarks. These results demonstrate the effectiveness of the hybrid token design in improving multimodal understanding without incurring additional costs.

5 Conclusion

In this work, we present Argus-Unified, a compact and economical UMM that integrates visual understanding and generation. Argus-Unified leverages pretrained VLMs to inherit strong visual priors and employs a hybrid token design. Extensive experiments demonstrate that Argus-Unified achieves state-of-the-art performance on multiple understanding benchmarks and competitive generation quality, while requiring the least amount of data and computation cost. We hope that this work can serve as a practical baseline for future research on developing UMM at a low cost.

Limitations

Argus-Unified focuses on the compact regime, and its effectiveness at larger model scales remains an open question. In addition, while our models demonstrate strong performance on multimodal understanding and generation benchmarks, the evaluation does not explore other capabilities such as image editing, which we leave for future work.

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A Implementation Details

In this section, we provide more details on the model architecture, datasets, and experiment setup.

A.1 Details on Model Architecture

Unified Vision Tokenizer. Our tokenizer consists of a vision encoder and a quantizer. We propose a hybrid token design that unifies the vision encoder by producing continuous token for image understanding and discrete token for image generation. The vision encoder is a vision transformer (ViT) (Dosovitskiy, 2020), specifically InternViT (Chen et al., 2024e). For Argus-Unified-1.5B, it is initialized from InternVL3-2B (Zhu et al., 2025), and for Argus-Unified-0.5B, from InternVL3-1B (Zhu et al., 2025). To remain consistent with the InternVL encoder, we split each image into n tiles of size $S \times S$ (where $S = 448$). The model size is approximately 300M parameters. We use $n = 6$ tiles for understanding and no tiling for image generation in Stage 2 training (Chen et al., 2024d). The encoder outputs continuous visual tokens, which follow two pathways: (1) they are directly fed into the LLM for image understanding tasks, and (2) they are passed to the quantizer to obtain discrete tokens for image generation. For quantization, we adopt a multi-codebook design inspired by UniTok (Ma et al., 2025a). We employ a lightweight transformer to infer multi-codebooks from LLM output visual tokens, as the final tokens input to the tokenizer decoder. Both continuous and discrete tokens are projected into the LLM’s feature space using separate MLP layers, one MLP layer for each token type.

Unified LLM. We use a standard decoder-only LLM to jointly train on mix image understanding and generation data. For Argus-Unified-1.5B, the LLM is Qwen2.5-1.5B (Qwen et al., 2024), initialized from InternVL3-2B (Zhu et al., 2025). For Argus-Unified-0.5B, the LLM is Qwen2.5-0.5B (Qwen et al., 2024), initialized from InternVL3-1B (Zhu et al., 2025).

Image Decoder. The image decoder is based on ViTamin-L/16 (Chen et al., 2024b), which combines transformer-based feature decoding with convolutional upsampling for spatial reconstruction. The decoder contains approximately 350M parameters.

Training Pipeline. As shown in Fig. 3, we train our models in two stages: unified vision tokenizer training and unified multimodal training. In Stage

1, we train the quantizer and the image decoder on top of the frozen vision encoder. In Stage 2, we train the LLM jointly with mixed understanding and generation data, while the image and text decoders are not involved in training. The LLM output text tokens are processed by the LLM linear head before computing the negative log-likelihood (NLL) loss. The LLM output visual tokens are further processed by a lightweight transformer head, similar to (Wu et al., 2024c; Ma et al., 2025a), before computing the NLL loss. The final training objective is the sum of these two losses.

Inference Pipeline. For image understanding, the input image is encoded by the vision encoder. The resulting continuous tokens are projected and concatenated with text tokens from the text encoder, and then fed into the LLM. The generated text tokens are decoded by the text decoder. For image generation, the text prompt is fed into the LLM to autoregressively predict discrete visual tokens, which are then decoded by the image decoder to produce the final image.

A.2 Details on Datasets

Tab. 7 summarizes the datasets used to train Argus-Unified. Stage 1 uses images from all the datasets, totaling 14M images. Stage 2 pretraining (PT) uses only ShareGPT4V (Chen et al., 2024c) pretraining subset and JourneyDB (Sun et al., 2023). Stage 2 supervised finetuning (SFT) uses 9.7M image-text pairs drawn from the following datasets: ShareGPT4V (Chen et al., 2024c), WiT (Srinivasan et al., 2021), VFLAN (Chen et al., 2024a), ScienceQA (Lu et al., 2022), and MGM-Instruct (Li et al., 2024d), JourneyDB (Sun et al., 2023), BLIP3-o (Chen et al., 2025a), and Echo-4o Instruct (Ye et al., 2025).

Besides the main training datasets, we also use additional datasets for ablation studies. To examine the impact of data volume in Stage 2 pretraining, we increase the data amount to 16.17M by incorporating additional datasets: a subset of MMC4 Core (3.8M) (Zhu et al., 2023), CC3M (2.9M) (Sharma et al., 2018), a subset of CC12M (2.9M) (Changpinyo et al., 2021), a subset of DALL-E 3 data (0.46M) (Egan et al., 2024), and a subset of DiffusionDB (0.76M) (Wang et al., 2022). These additional datasets sum to 10.82M. Combined with ShareGPT4V PT (1.25M) (Chen et al., 2024c) and JourneyDB (4.1M) (Sun et al., 2023), the total pretraining data volume is 16.17M.

In addition, we conduct ablation on the Stage 1

Table 7: Summary of datasets used in Argus-Unified. ✓ indicates that the dataset is used in the corresponding training stage. Note that Stage 1 uses only the images from these datasets.

Datasets	Size (M)	Stage 1	Stage 2 PT	Stage 2 SFT
ShareGPT4V PT (Chen et al., 2024c)	1.25	✓	✓	
ShareGPT4V SFT (Chen et al., 2024c)	0.67	✓		✓
WiT (Srinivasan et al., 2021)	0.54	✓		✓
VFLAN (Chen et al., 2024a)	1.50	✓		✓
ScienceQA (Lu et al., 2022)	0.01	✓		✓
MGM-Instruct (Li et al., 2024d)	1.50	✓		✓
JourneyDB (Sun et al., 2023)	4.10	✓	✓	✓
BLIP3-o (Chen et al., 2025a)	0.06	✓		✓
Echo-4o Instruct (Ye et al., 2025)	0.07	✓		✓
CC3M (Sharma et al., 2018)	2.92	✓		
DALL-E 3 (Egan et al., 2024)	1.01	✓		
DiffusionDB (Wang et al., 2022)	2.00	✓		
Total	15.6			

Table 8: Comparison with understanding-only models. Argus-Unified achieves superior performance on GQA (Hudson and Manning, 2019), POPE (Li et al., 2023), and VQAv2 (Antol et al., 2015; Goyal et al., 2017), while remaining competitive on MME-P (Zhang et al., 2021).

Model	LLM Scale	GQA↑	MME-P↑	POPE↑	VQAv2↑
MobileVLM (Chu et al., 2023)	1.4B	56.1	1196.2	84.5	-
MobileVLM-V2 (Chu et al., 2024)	1.4B	59.3	1302.8	84.3	-
MiniGemini (Li et al., 2024e)	2B	59.9	1341.0	83.9	-
MobileVLM (Chu et al., 2023)	2.7B	59.0	1288.9	84.9	-
MobileVLM-V2 (Chu et al., 2024)	2.7B	61.1	1440.5	84.7	-
LLaVA-Phi (Zhu et al., 2024)	2.7B	-	1335.1	85.0	71.4
LLaVA (Liu et al., 2023)	7B	-	809.6	76.3	-
LLaVA-v1.5 (Liu et al., 2024b)	7B	62.0	1510.7	85.9	78.5
InstructBLIP (Dai et al., 2023)	7B	49.2	-	-	-
Qwen-VL-Chat (Bai et al., 2023)	7B	57.5	1487.5	-	78.2
IDEFICS-9B (Laurençon et al., 2023)	8B	38.4	-	-	50.9
InstructBLIP (Dai et al., 2023)	13B	49.5	1212.8	78.9	-
Argus-Unified	0.5B	61.2	1323.2	87.7	77.4
Argus-Unified	1.5B	63.0	1405.1	87.7	79.1

training data by using 51.2M of DataComp (Gadre et al., 2023). More details are provided in Sec. B.

A.3 Details on Experiment Setup

All the experiments are conducted in one node of 8 NVIDIA H100 GPUs. By using the pretrained InternViT from InternVL3 (Zhu et al., 2025), we follow similar image preprocessing to be consistent with the encoder design for image understanding. In particular, an input image is first resized to the closest aspect ratio from a predefined set. The image is then partitioned into n square tiles, each of which is further resized to match the input resolution $S \times S$ of the vision transformer. The original input image is also resized to the same resolution,

regarded as a thumbnail. Each tile and the thumbnail are independently encoded by the ViT into sequences of visual patch tokens. To reduce the total number of tokens, features from spatially adjacent patches are concatenated along the channel dimension. In our experiments, we use the same resolution $S = 448$ as InternVL3 (Zhu et al., 2025) vision encoder. We set $n = 6$ tiles for image understanding and no tiling for image generation.

By default, Stage 1 training is conducted for two epochs on 14M images with a batch size of 128. Stage 2 uses a batch size of 32 and a learning rate of 5×10^{-5} during pretraining, and a batch size of 64 with a learning rate of 1×10^{-4} during supervised finetuning. Both pretraining and supervised

Table 9: Comparison with generation-only models. Argus-Unified achieves superior performance on MJHQ-30K (Li et al., 2024a) and GenEval (Ghosh et al., 2023) benchmarks.

Models	Type	Model Scale	MJHQ-30K FID↓	GenEval Overall↑
LlamaGen (Sun et al., 2024)	AR	0.8B	25.6	0.32
LDM (Rombach et al., 2022)	Diffusion	0.4B	-	0.37
SD v1.5 (Rombach et al., 2022)	Diffusion	1B	-	0.43
PixArt- α (Chen et al., 2023)	Diffusion	0.6B	-	0.48
SD v2.1 (Rombach et al., 2022)	Diffusion	1B	27.0	0.50
SD-XL (Podell et al., 2023)	Diffusion	2.6B	8.8	0.55
DALL-E 3 (Betker et al., 2023)	Diffusion	-	-	0.67
Argus-Unified	AR	0.5B	8.0	0.66
Argus-Unified	AR	1.5B	6.9	0.71

Table 10: Comparison of data volume and type in unified multimodal training with a pretrained VLM. More supervised finetuning (SFT) data improves performance (row 1 vs 2), while excessive pretraining (PT) data (row 3) or generation-only PT data (row 4) degrades performance. These results highlight the insights of constructing data in unified multimodal training with pretrained VLMs.

Datasets (Volume)		Image Understanding				Image Generation	
PT	SFT	GQA ↑	MME-P ↑	POPE ↑	VQAv2 ↑	MJHQ-30K ↓	GenEval ↑
und+gen (5.35M)	und+gen (4.77M)	62.0	1283.1	87.8	77.6	8.7	0.30
und+gen (5.35M)	und+gen (9.58M)	63.0	1405.1	87.7	79.1	8.2	0.71
und+gen (16.17M)	und+gen (9.58M)	62.5	1314.9	88.1	78.6	8.0	0.68
gen (4.1M)	und+gen (9.58M)	62.6	1379.1	87.1	78.8	8.3	0.68

finetuning are run with 1 epoch.

B More Experiment Results

Comparison with Existing Understanding-only Models. We further compare Argus-Unified with vision–language models designed solely for image understanding, including the MobileVLM series (Chu et al., 2023, 2024) and the LLaVA series (Liu et al., 2023, 2024b; Zhu et al., 2024), as shown in Tab. 8. Argus-Unified-1.5B achieves the best performance on GQA, POPE, and VQAv2 benchmarks among models with LLM scales ranging from 1.4B to 13B. Moreover, compared with MobileVLM and MobileVLM-V2 (both using a 1.4B LLM), even our smallest variant (0.5B LLM) delivers substantially stronger performance. These results highlight the strong image-understanding ability of Argus-Unified.

Comparison with Existing Generation-only Models. We also compare Argus-Unified with representative image generation models, including Stable Diffusion models (Rombach et al., 2022; Podell et al., 2023), DALL-E 3 (Betker et al., 2023), and the autoregressive model LlamaGen (Sun et al., 2024). As shown in Tab. 9,

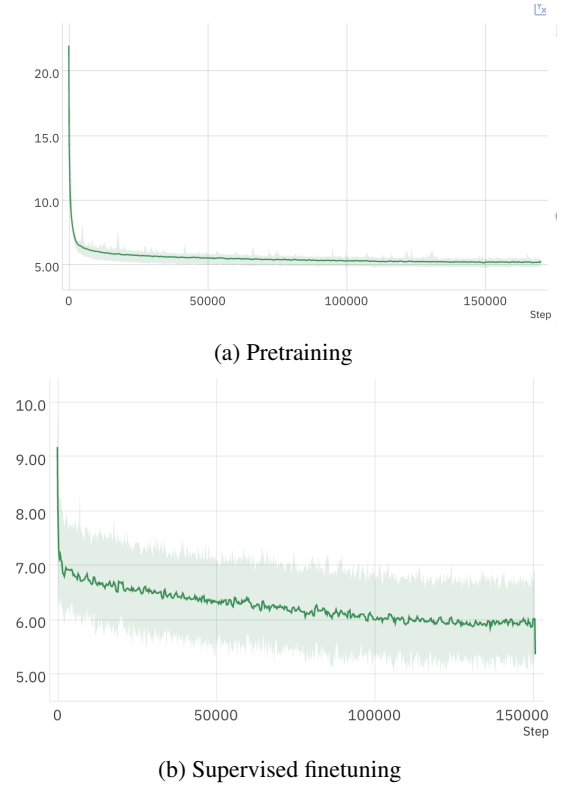


Figure 6: Loss curves over the course of Stage 2 training.

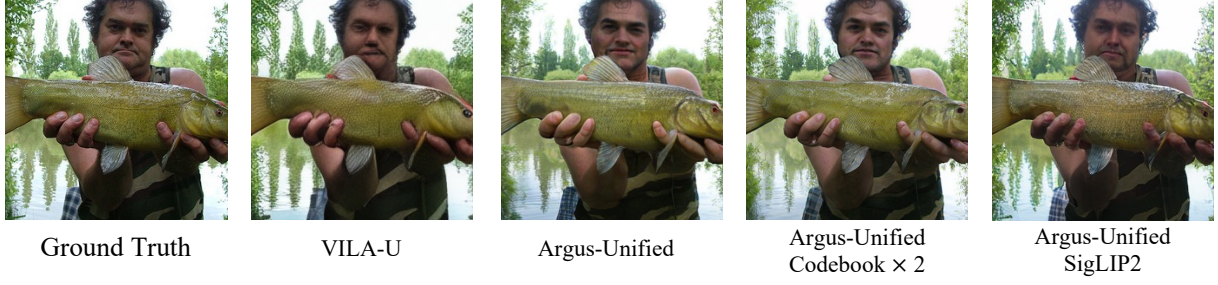


Figure 7: Comparison of reconstructed images on ImageNet. Our frozen-encoder design produces high-fidelity reconstructions and seamlessly generalizes to larger codebook and alternative vision encoders such as SigLIP2 (Tschannen et al., 2025).

Table 11: Ablation of Stage-1 tokenizer training data. Training the tokenizer on 51.2M of DataComp (Gadre et al., 2023) images improves ImageNet reconstruction (i.e., lower rFID) but yields weaker unified model performance than our mixed 14M data, indicating that reconstruction quality does not directly predict generation or understanding performance. It also highlights the effectiveness of our dataset curation for Stage 1 training.

Stage 1 Data	Image Reconstruction	Image Understanding				Image Generation	
	ImageNet ↓	GQA ↑	MME-P ↑	POPE ↑	VQAv2 ↑	MJHQ-30K ↓	GenEval ↑
Mixed-14M	1.60	62.95	1405.06	87.70	79.10	8.22	0.71
DataComp-51.2M	1.15	62.37	1384.09	87.47	79.29	8.68	0.69

Table 12: Comparison of image generation output resolutions. Higher resolution (448×448) improves both image understanding and generation performance compared to 256×256 .

Output Resolution	Image Understanding				Image Generation	
	GQA ↑	MME-P ↑	POPE ↑	VQAv2 ↑	MJHQ-30K ↓	GenEval ↑
256×256	62.65	1365.53	87.22	79.25	8.58	0.71
448×448	62.95	1405.06	87.70	79.10	8.22	0.71

Argus-Unified-1.5B achieves an overall accuracy of 71% on GenEval (Ghosh et al., 2023) and an FID of 6.9 on MJHQ-30K (Li et al., 2024a), outperforming these popular generation models on both benchmarks. Our smaller variant (0.5B) also delivers notably strong results, demonstrating the high quality of images produced by Argus-Unified and the effectiveness of our training pipeline.

Data Volume and Data Type. We further investigate how to better utilize data when leveraging a pretrained VLM for UMM. Prior works typically adopt larger data volume for pretraining (PT) than for supervised fine-tuning (SFT) (Wu et al., 2024a; Ma et al., 2025a). However, this trend does not hold when initializing from a pretrained VLM. As shown in Tab. 10, increasing the amount of SFT data improves performance (row 1 vs. row 2), but enlarging the PT data size from roughly 5M to 16M degrades performance on most benchmarks (row 2 vs. row 3). This suggests that the pretrained VLM already provides strong visual priors, and

PT mainly serves as a warm-up phase for adapting to visual token generation. Moreover, we further study whether understanding data is needed by using only generation data in PT. This leads to consistently worse performance across all benchmarks (row 2 vs. row 4). These findings highlight the insights in constructing data when leveraging pretrained VLMs.

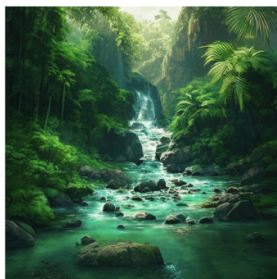
Loss Curve Over the Course of Training. Fig. 6 shows the loss curves for Stage 2 pretraining and supervised finetuning over the course of training. In both stages, the loss continues to decrease steadily throughout training. It demonstrates that our training is stable and well-behaved across both pretraining and finetuning.

Vision Tokenizer. We further conduct ablation study on unified vision tokenizer training by training another tokenizer using 51.2M images from DataComp-1B (Gadre et al., 2023) (where UniTok (Ma et al., 2025a) trains on the full dataset). As shown in Tab. 11, although using 51.2M data

achieves a lower ImageNet reconstruction rFID, using it for unified training leads to comparable but slightly worse performance compared to our default tokenizer trained on mixed 14M set. All Stage-2 training settings are exactly the same. This indicates shows that better reconstruction quality does not necessarily translate into stronger generation ability, and highlights the effectiveness of our dataset construction for Stage 1 unified vision tokenizer training.

Generation Output Resolution. Our framework can flexibly support different image generation output resolutions. We compare the performance of using resolution 256×256 and 448×448 in Tab. 12. With strong performance obtained with resolution 256×256 , using higher resolution can further increase the performance on both image understanding and image generation benchmarks. By default, we use the output resolution of 448×448 in our experiments.

More Image Generation Results. To complement the visual examples in Fig. 5, we additionally include their corresponding prompts and more generated samples in Fig. 8.



Amazon rainforest river



Area of Effect Healing tree, realism, welcoming, marketable



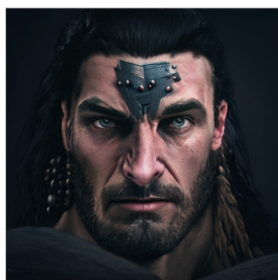
Man Monkey with headphone, singer, cyberpunk, lens 85mm



Miniature clay burger



A watercolor painting of pink roses, in the style of Thomas W Schaller, in the style of serene watercolors, loose handling of paint



A headshot of an ancient Greek warrior looking from under the brows ready to fight, black hair, dark blurry background



Ford Mustang, by Kim Jung Gi, bold Colors, semi realistic, glossy, digital art, white background



A majestic Goddess of beauty, charming dressed in a regal, jeweled gown and ornate crown, her golden hair cascading down her back, in the style of Pino Daeni



A kelp forest in a deep ravine



A snow mountain



German shepherd dressed as a general in army



Assorted drawings of flowers, korean style, pop art, flat, vectorized



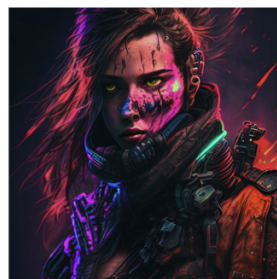
Professional studio photo of elderflowers, photo taken from straight above at 90 degrees on white background FFFFFF



Happy dreamy owl monster sitting on a tree branch, colorful glittering particles, forest background, detailed feathers



Exterior render of an extremely luxurious villa with glass and pastel walls, sharp focus, cinematic, award winning photography, high contrast, sharp focus, HD cinematic lighting



Hyper realistic portrait photo, battle worn female, heavily worn damaged tactical street gear, dramatic scar across missing eye, warm contrasting neon colour

Figure 8: Images generated by Argus-Unified from the corresponding text prompts.